Principles of Scientific Computing

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Welcome

Principles of Scientific Computing (psc) collects practical guidance on writing, organizing, sharing, and maintaining code for research projects. It is the computing companion to *Principles of Scientific Writing*.

The guidance applies to any research project, whether it is a lab analysis, a course, or a software package. Most examples use R, but the principles carry over to other languages.

What's here

- **Writing code**: coding style, coding practices, data masking, and working with big data.
- **Projects**: code repositories, reproducible environments, and code publication.
- **Collaboration**: Git and GitHub, collaborative software development, and continuous integration.
- **Tools**: the Unix command line and Quarto.

Each page is also available as RevealJS slides (`-slides.html`), a PDF handout (`-handout.pdf`), and a Word document (`.docx`).

Related guides

- *Principles of Scientific Writing* covers writing about research and statistical analysis.
- *Working with AI* covers AI coding assistants and agent-driven pull requests.
- The UCD-SERG lab manual covers policies and resources specific to the UC Davis Seroepidemiology Research Group, such as computing clusters, onboarding, and communication.

Origin

These chapters began in the UCD-SERG lab manual and moved here, with their history, so that they can serve projects outside that group.